

TECHNICAL ROADMAP TEMPLATE



TEAM INFORMATION

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Institution	Bina Nusantara University
Country	Indonesia
Track	Climate Change Resilience
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SECTION 1: EXECUTIVE SUMMARY (PROBLEM-SOLUTION FIT)

Southeast Asia faces an escalating flood crisis driven by climate change. Data from ADINet (AHA Centre) shows floods account for approximately 67% of all regional disasters since 2012. In Indonesia, BNPB data reveals flood events rose from 1,420 cases in 2024 to 1,652 in 2025 (a ~16% increase), with 7,650 flood incidents recorded over the past five years. Countries including the Philippines, Indonesia, Vietnam, Thailand, and Malaysia are classified as “very high” flood risk by the World Risk Report 2025.

Despite this severity, communities lack timely, actionable evacuation guidance. Existing early warning systems provide alerts but fail to translate them into clear actions or safe routes, increasing risk and casualties.

Escood addresses this gap through a Localized Hybrid Intelligence System powered by a Multi-modal AI Framework. The system operates in three stages: (1) Predictive Analytics for early risk detection, (2) Intelligent Validation to filter misinformation via community-sensor cross-referencing, and (3) Adaptive Response for real-time emergency guidance. Critically, Escood uses an Offline-First Architecture with On-Device Proactive Risk-Caching, integrating Digital Elevation Models (DEM) locally to generate adaptive routing to Safe Zones even without internet connectivity. This transforms fragmented disaster data into a reliable Single Operational Picture, ensuring life-saving intelligence remains accessible regardless of infrastructure disruption.

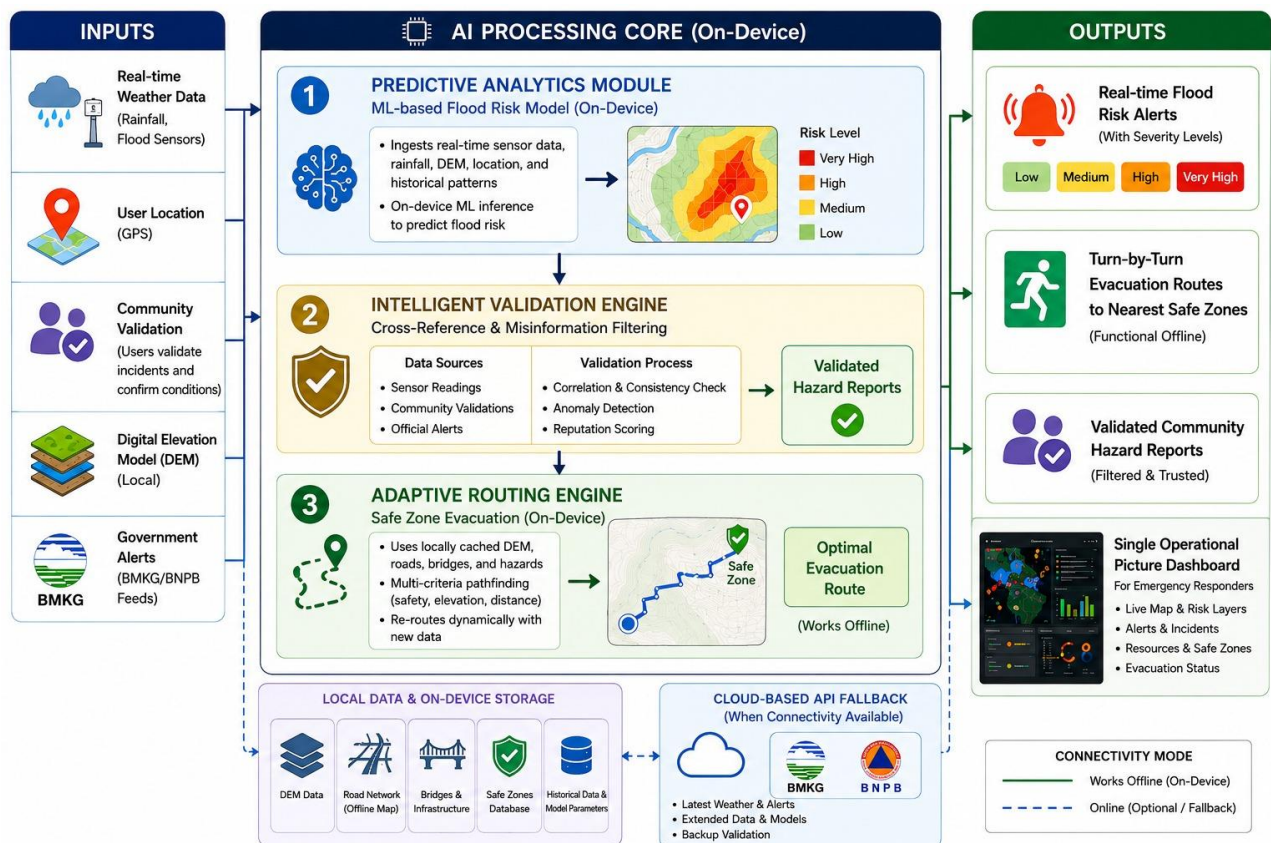
SECTION 2: TECHNICAL ARCHITECTURE

Provide a high-level overview of how your AI system will work. You are encouraged to include a

2.1 System Components:

- **Inputs:** Real-time weather data (rainfall, flood sensors), user location (GPS), community-reported incidents, Digital Elevation Model (DEM) data (stored locally), and government BMKG/BNPB flood alert feeds.
- **Processing Core:** A three-layer AI pipeline: (1) Predictive Analytics module using an ML-based flood risk model running on-device, (2) Intelligent Validation engine cross-referencing sensor data with community reports to filter misinformation, and (3) Adaptive Routing engine leveraging locally cached DEM data to compute optimal Safe Zone evacuation paths. Cloud-based API fallback is used when connectivity is available.
- **Outputs:** Real-time flood risk alerts with severity levels, turn-by-turn evacuation routes to nearest Safe Zones (functional offline), validated community hazard reports, and a unified Single Operational Picture dashboard for emergency responders.

2.2 Architecture Diagram:



SECTION 3: AI APPROACH & MODEL SELECTION

Identify the specific AI techniques and tools you plan to use.

- **Primary AI Approach:** [X] Machine Learning [] NLP [] Computer Vision [] Generative AI [X]
Other: Multi-modal Hybrid AI (ML + Rule-based Validation)
- **Model Selection:** TensorFlow Lite for on-device flood risk prediction (lightweight ML); scikit-learn for data preprocessing and validation logic; OpenStreetMap + DEM data with custom routing algorithm for Safe Zone pathfinding; mobile framework for cross-platform app deployment.

- **Reasoning:** TensorFlow Lite is chosen for its proven performance on low-resource mobile devices, critical for Indonesia's infrastructure context. The offline-first architecture with on-device DEM caching ensures functionality during connectivity disruptions—the exact scenario when evacuation guidance is most needed. The hybrid approach combining ML prediction with rule-based validation provides high-integrity outputs by cross-referencing multiple data sources to reduce misinformation during emergencies.

SECTION 4: DATA STRATEGY & ETHICS

(Based on Workshop 2: Technical Architecture & Data Ethics)

- **Data Sources:** Open-source datasets: BNPB (Indonesia National Disaster Management Agency) flood records, BMKG weather data, AHA Centre ADINet regional disaster database, NASA/USGS SRTM Digital Elevation Model (DEM), OpenStreetMap for road and infrastructure data, and community-submitted incident reports via the app.
- **Data Quality & Cleaning:** Data will be validated through the Intelligent Validation layer, cross-referencing sensor readings against community reports and official feeds to detect and filter anomalies. Historical flood data will be cleaned using Python (pandas) with outlier removal, normalization, and handling of missing values. Community-submitted reports will be flagged for review if they contradict verified sensor data.
- **Licensing & Legality:** Yes. All primary data sources are publicly licensed: BNPB and BMKG data are open government datasets (public domain), NASA SRTM DEM is freely available for non-commercial and research use, OpenStreetMap uses the Open Database License (ODbL), and ADINet data is openly accessible for humanitarian purposes. All sources are compliant with hackathon usage requirements.
- **Bias Mitigation & Fairness:** Training data is sourced from diverse geographic regions across Indonesia to prevent urban-rural bias. The app interface is designed in Bahasa Indonesia with simple visual cues to reduce linguistic and literacy barriers. Regional flood patterns from all 34 provinces are included to avoid geographic bias. Community report validation prevents over-representation of tech-savvy urban users by weighting verified sensor data equally with crowdsourced inputs.

SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

(Based on Workshop 3: Agile Prototyping)

Outline your development sprints leading up to the Semi-Finals in late June.

Phase	Activity / Task	Tools Used	Expected Outcome

Sprint 1 (May 1-15)	Data collection & Environment Setup	GitHub, Python	Cleaned Dataset
Sprint 2 (May 16-31)	MVP / Core AI Model Development	VS Code, Cursor	Basic functional AI
Sprint 3 (June 1-15)	UI/UX & Integration	Figma, React/Flask	Working Prototype
Sprint 4 (June 16-25)	Testing, Debugging & Pitch Recording	GitKraken, Loom	Final Demo Video

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

How can your solution expand beyond your local university or city?

- **Scaling Potential:** Escood's modular architecture allows rapid adaptation across ASEAN countries. The DEM and flood data layers can be replaced with region-specific datasets (e.g., Philippines PHIVOLCS, Vietnam MONRE, Thailand TMD) without changing the core AI pipeline. The offline-first design is particularly valuable for rural areas in Myanmar, Laos, and Cambodia with limited connectivity. Multi-language support can be added for Thai, Vietnamese, Filipino, and Malay, enabling ASEAN-wide deployment within 6–12 months post-prototype.
- **Technical Constraints:** Current limitations include: (1) DEM caching increases local storage requirements (~500MB per region), which may be a constraint on low-end devices; (2) The ML flood prediction model requires initial calibration with local hydrological data for accuracy in new regions; (3) Real-time community validation depends on sufficient user adoption to generate meaningful crowdsourced data; (4) Integration with official government sensor networks (BNPB/BMKG APIs) requires formal data-sharing agreements for production deployment.